



Automating UNIX Administration

A Puppet show can turn out to be real entertainment for UNIX administrators.

In the UNIX operating system everything is a file, which makes it an easy-to-manage and administrator-friendly system. The traditional way of managing UNIX was to use the telnet interface, but being a plain-text protocol, telnet exposes you to the risk of network snooping and compromise of login credentials. SSH works on an encrypted channel to overcome the snooping issues. A UNIX administrator can SSH into the box from a remote machine and change the configuration or execute commands remotely.

Generally, it is considered a good practice to take a configuration backup before making any changes to the production configuration so that the old configuration is available for roll-back. Also, as a part of the organisation's policy, the same base configuration should be configured on all the servers to reflect consistency and as a server-hardening practice. A majority of the

problems in the UNIX environment occur due to ad-hoc changes, which can be mitigated by following proper change management procedures. Handling and monitoring ad-hoc changes, and restoring the previous state, remains a challenge for organisations.

Meeting such challenges is quite workable for a small set-up of 1-20 servers and a dedicated UNIX administration. But during hardware failure or other problems, where the servers need to be reconfigured from scratch, it takes a lot of effort and time in restoring the servers to the previous state. To handle such scenarios, a quick solution would be to hire another UNIX administrator who could act as a secondary resource and offloads other activities from the primary resource during disaster conditions.

Think about a scenario of managing a globally-distributed data centre with 500 *NIX servers or more, comprising Solaris, Debian, Ubuntu, Fedora, CentOS, etc. Here, servers